Date.....
M.....
Y 20.....

Page.....

Research project periodic log entry form

Project ID : R26-SE-039
Student ID : IT22251114
Name : Jinad Induvidwa A.G
Individual component : AI driven user story generation

Summery of R.P work carried out during the reporting period.

Date	RP work
01/04/2026	conduct initial research and identify suitable technologies (LLMs, speech to text models)
03/04/2026	planed overall system architecture and design. the project structure
05/04/2026	started the backend development - integrate pyAudio and implement real time voice to text conversion using medium-en model
10/04/2026	improve backend pipeline to process and store transcription .txt file for further processing
13/04/2026	develop frontend UI using dummy data and implement responsive desing

Supervisor's comments: at this stage the student was expanding the scope
Student processing well in research project : (yes)/no
If no :
project work to be carried out before next evolution period:
code base checked : yes/no

signature :
date : 20/04/2026

Research project periodic log entry form

Project ID : R26-SE-039
Student ID : IT22251114
Name : Jinad Induvidua AG.
Individual component : AI Driven user story generator and conflict detector

Summery of RP work carried out during the reporting period.

Date	RP work
16/04/2026	capture raw browser microphone buffer and downsampled them dynamically from native rate 16Hz
20/04/2026	design a lightweight, bidirectional web gateway in FastAPI to stream high-frequency binary PCM array
23/04/2026	integrate azure cognitive service SDK to continuously feed the active PCM stream into a non-blocking speech recognizer using asynchronous event handler.
27/04/2026	constructed thread-safe local session store in FAST API to manage concurrent active connection

Supervisors comment :

Student processing well in research project : Yes / NO
If NO :

Project work to be carred out before next evalution period :

codebase checked :

Signature : ~~Jin~~
and :
Date : 31/07/2026

Research project periodic log entry form

Project ID : R26-SE-039
Student ID : IT22251114
Name : Jinad Induwara A.G
Individual component : AI driven user story generator and conflict detector

Summary of RP work and carried out during the reporting period

Date	RP work
03/05/2026	Built a cleanup worker in the speech persistence layer that auto finalize transcripts as soon as all active participants disconnected.
08/05/2026	developed regex-based filters to strip conversational filler from transcripts while handling multi-line continus speech matching
11/05/2026	Engineered a specialized sliding window chunker that groups dialogue by speaker boundries
13/05/2026	Implement local vector index using chrom DB, supporting high-dimensional OpenAI embeddings

Supervisor's comments:

Student processing well in research project : Yes / NO
If NO :

Project work to be carried out before next evolution period:

code base checked : Yes / NO

Signature and : [signature]
Date 31/07/2026

Research Project Periodic log entry form

Project Id : R26-SE-039
Student Id : IT22251114
Name : Jinad Induvidwa A.G
Individual component : AI Driven user story generator

Summary of RP work carried out during the reporting period

Date	RP work
16/05/2026	Build transactional pipeline that drives ingestion, vector indexing, retrival, generation and validation
20/05/2026	Design custom system prompts and restrict the LLM retrived context chunks forcing it to every agile requirement to specific transcript source ID's
27/05/2026	Implement automated audit validator that parses generated stories, checks for structural format ("As a"), validate Given-When-Then rules. and alerts users of unsupported claims.

Supervisor's comments:

Student process well in research project: Yes / No
If NO :

Project work to be carried out before next evolution period:

codebase checked : Yes/ No

Signature and date : 31/07/2026

Research project periodic log entry form

Project Id : R26-SE-039
Student Id : IT22261114
Name : Jinad Induvidwa A.G
Individual Component: AI driven user story generator and conflict detector

Summary of RP work carried out during the reporting period

Date	RP work
03/06/2026	Design the meeting requirement thread manager, introducing requirement life cycle state
07/06/2026	Improved requirement extraction by filtering out non-requirement utterances before RAG pipeline, reducing unnecessary LLM process
10/06/2026	Design requirement memory layer using PG SQL with pgvector to maintain contextual history of requirement across the meeting instead of relying only on temporary LLM context

Supervisor's comment:

Student processing well in research project : Yes / No
If NO:

project work to be carried out before next evaluation period:

Codebase checked : Yes / No

Signature :
Date : 31/07/2026

Research Project Periodic log entry form

Project ID : R26-6E-039
Student ID : IT22261114
Name : Jined Induwidwa A.G.
Individual component : AI Driven user story generator and conflict detector.

Summery of RP work carried out during the reporting Period.

Date	RP work
16/06/2026	Redesign the knowledge base service as a seperate FAST API microservices to support feature multi-tenant Saas Application.
18/06/2026	evaluated embeddings strategies and finalize using gemini embeddings for semantic retrival while planing to use.
20/06/2026	improved the user story validation workflow by introducing semantic similarity validation rather then relying only reward matching.
28/06/2026	enhanced the system Architecture to support requirement conflict detection, enabling identification of contradiction stakeholder requirement during live discussion.

Supervisor's comment :

Student processing well in research project : Yes / No
If NO :

Project work to be carried out before Next evalution period :

Codebase checked : Yes / No

Signature and Date : 31/07/2026.

Research Project Periodic log entry form

project ID : R-26-SE-030
Student ID : IT22251114
Name : Jinod Induvidu A.G
Individual component : AI Driven user story generator and conflict detector

Summary of RP work carried out during the reporting period.

Date	RP work.
02/07/2026	completed the implementation of the indic voice parser service using fast-whisper, enabling real-time speech transcription, through a FAST API backend.
04/07/2026	Developed the frontend meeting dashboard with support for microphone control, screen sharing, camera controls and live transcript visualization.
07/07/2026	Design and implemented the database initialization strategy for backend service, evaluating SQL initialization.

Supervisor comment :

Student processing well in research project: Yes / NO
If NO :

project work to be carried out before next evolution period :

code base checked : Yes / NO

Signature and : Jina
Date 31/07/2026

Research project periodic log entry form

Project ID: R26-SE-030
Student ID: IT22251114
Name : Jinad Induwara A.G
Individual component: AI Driven user story generator and conflict detector

Summary of RP work carried out during the reporting period.

Date	RP work
16/07/2026	enhance the RAG retrival strategy researching solution to reducing top-K retrival limitation and improve semantic document section
20/07/2026	Finalized the integration plan for gemini powered semantic processing, ensuring compatibility between retrival, requirement validation and user story generation component.
25/07/2026	revised and refined the backend API implementation for speech-to-text services, improving service organization, routing and preparation for integrating diarization

Supervisor comment :

Student processing well in research project: Yes/NO
If NO:

project work to be carried out before next evalution period

codebase checked: Yes/NO

signature : Jin
and
Date 31/07/2026

Research Project Periodic log entry Form

Project ID: R26-SE-0301
Student ID : IT22251114
Name : Jinad Induvidua A.G
Individual component: AI Driven user story generator and conflict detector.

Summary of RP work carried out during the reporting period

Date	RP work.
02/08/2026	Designed and execute unit test suit for individual microservice modules
05/08/2026	Create integration test to validate end to end communication and API contract between frontend, API gateway, Auth and test case generation
10/08/2026	conduct performance and stress testing under simulated concurrent user traffic to measure system throughput, API latency and LLM generation response time

Supervisor comment:

Student Processing well in research project : Yes/NO
If NO :

Project work to be carried out before Next evolution period

codebase checked : Yes/NO

Signature : Jin
and :
Date 31/07/2026

Research project periodic log entry form

Project ID : R26-SE-301
Student ID : IT22251114
Name : Jinad Induduwa A.O
Individual component : AI Driven user story generation and conflict detector

Summery of RP work carried out during the reporting period.

Date	RP work.
17/08/2026	configured and launched an AWS EC2 instance using Amazon linux 2023 Provide a cloud-based envoronment
23/08/2026	containerized the application component using docker to ensure consistent deployement
25/08/2026	configre docker volume to persist application and model related data
28/08/2026	Deploye the recived application service on EC2 instance and configure CI/CD Pipeline.

Supervior comment

student processing well in research project : Yes/No

Project work to be carried out before Next evolution period

codebase checked : (Yes)/NO

Signature and Date : 31/08/2026

Research Project Periodic log entry form

Project ID : R26-SE-039
Student ID : IT22251114
Name : Jinad Induwara AG
Individual component : AI driven user story generator and conflict detector

Summery OF RP work and carried out during the reporting period

Date	RP work
03/09/2026	Defined evaluation metrics and prepared a structured feedback questionnaire
10/09/2026	Identified and reached out to target group of evaluators
13/09/2026	conduct evaluator session where participants generated user stories and tested core system workflow on deployed application

Supervisor's comment

Student processing well in research project : Yes/NO
If NO :

project work to be carried out before next evalution period

codebase checked : Yes/NO

Signature : [signature]
and Date : 18/09/2026

Research Project Periodic log entry form

Project ID : R26-SE-039
Student ID : IT22251114
Name : Jinad Induwara A.G
Individual component : AI driven user story generator and conflict detector

Summery of RP work and carried out during the reporting period

Date	RP work.
16/09/2026	collected completed survey responses, system usability scale scores and qulitative notes regarding generate accuracy, prompt handling and edge cases.
21/09/2026	Documented evaluation finding and Prioritized recomended changes into immidiate UI/API refinement

Supervisor's comment :

Student Processing well in research project : Yes/NO
If NO :

Project work to be carred out before next evalution period :

codebase checked : Yes/NO

Signature Jin
and
Date. 30/09/2026

Research project Periodic log entry form

Project ID : R26-SE-039
Student ID : IT22291114
Name : Jinad Induvidwa AG
Individual component : AI driven user story generator and conflict detector

Summary of RP work and carried out during the reporting period

Date	RP work
02/10/2026	Authored core chapters of the individual thesis.
04/10/2026	Authored detailing the Problem domain, component Architecture, implementation details.
10/10/2026	formatted the individual thesis according to university academic guidlines

Supervisor comments :

Student processing well in research project : Yes/NO
If NO :

project work to be carried out before next evalution period :

code base checked : Yes/NO

Signature and :

Date 18/10/2026

Research project periodic log entry form

Project ID : R26- SE - 039
Student ID : IT22251114
Name : Jinad Indavidwa A.G
Individual component : AI driven user story generator and conflict detector

Summary of RP work and carried out during the reporting period

Date	RP work
16/10/2026	collaborated with team members to merge individual component section in to the comprehensive group thesis.
18/10/2026	conduct comprehensive team review of the final group report to ensure cross chapter consistency, uniform formatting, synchronized terminology

Supervisor comments :

Student Processing well in research project : Yes / No

Project work to be carried out before next evalution period :

code base checked : Yes / No

Signature and :

Date 19/10/2026